Hardware-Aware Training and Precision-Retention Frontiers for Low-Precision Analog Vision

Songqiao Li

Abstract

We present a behavioral simulation framework for analog compute-in-memory inference, focusing on precision-retention trade-offs under realistic physical deployment constraints. Under a tested AIHWKit IdealDevice noise baseline, standard fixed-instance hardware-aware training achieves 87.28% fresh-instance accuracy at 8-bit precision but collapses to 14.64% at 4-bit. We show that Ensemble Hardware-Aware Training (HAT), which resamples device-to-device mismatch masks during training, rescues the 4-bit pure-quantization regime, recovering fresh-instance accuracy to 86.16%. We then evaluate a realistic phase-change memory (PCM) simulation regime. Under the tested PCM UnitCell, models converge across 4-bit, 6-bit, and 8-bit precisions, but retention behavior is precision dependent. The results reveal a precision-retention deployment frontier: 8-bit PCM is drift-flat over 24 hours, while 4-bit PCM reaches the most aggressive tested compression but incurs a measurable 4.01 pp time-dependent drift. In this tested sweep, 6-bit PCM is the best observed Pareto midpoint, maintaining 8-bit-like drift stability while achieving comparable fresh accuracy. These findings separate the algorithmic challenge of cross-instance robustness from the physical constraints of analog retention.

Keywords: analog compute-in-memory; hardware-aware training; phase-change memory; low-precision inference; precision-retention tradeoff; vision transformers; mixed-signal simulation.

1 Introduction

Compute-in-memory (CIM) architectures execute dense matrix-vector multiplications directly within memory arrays, eliminating the data-movement bottleneck that limits conventional digital systems[1–3]. Across emerging analog memories, including resistive memories, phase-change memory (PCM), and organic optoelectronic devices, deployment accuracy is shaped by finite conductance precision, device-to-device variability, cycle-to-cycle noise, and retention drift[4–8]. These effects are often characterized at the device level, but their system-level consequences for modern vision backbones remain difficult to rank before fabrication.

However, translating device-level models into deployment-facing model accuracy faces a critical methodological gap. Mature CIM simulators expose increasingly detailed analog effects[2;4;5;9], yet the algorithmic and physical failure modes are often entangled: a low-precision model can fail because hardware-aware training overfits one fixed mismatch realization, because quantization is too coarse, or because a realistic memory substrate drifts after programming. Device-specific studies, including organic and optoelectronic demonstrations, further highlight this gap because variability, ADC resolution, retention-induced drift, and transfer across fabricated instances are frequently absent from network-level analyses or treated only qualitatively[10–12]. A deployment workflow therefore needs to separate algorithmic robustness from physical precision-retention limits.

Here we present a behavioral simulation framework that evaluates analog neural network deployment under both idealized pure-quantization and realistic device-physics regimes. We apply a mixed-signal deployment model to the Tiny-ViT-5M backbone to systematically disentangle algorithmic generalization failures from physical retention limits. Three concrete contributions follow. First, we expose a fresh-instance transfer failure in standard fixed-instance hardware-aware training at extreme low precision (4-bit), where accuracy collapses to near-chance. We introduce Ensemble Hardware-Aware Training (HAT), which resamples device-to-device mismatch masks each epoch, rescuing the 4-bit pure-quantization regime and restoring fresh-instance accuracy to 86.16±0.19%. Second, we translate this algorithmic rescue into a realistic phase-change memory (PCM) simulation regime and observe stable convergence under the tested PCM UnitCell setting where pure quantization fails. Third, we map a 4/6/8-bit precision-retention deployment frontier for PCM. In this tested sweep, 4-bit is trainable but drift-limited, 8-bit is drift-flat, and 6-bit provides the best observed Pareto midpoint between model compression and long-term inference reliability.

The framework is intentionally behavioral rather than circuit-accurate, focusing on macroscopic precision and drift phenomena rather than array-level parasitics (IR drop, sneak paths) which remain crucial future considerations. Additional supporting analyses, including front-end compensation and zero-shot transfer to alternative organic optoelectronic profiles, are provided in the Supplementary Information. Its purpose is to provide a reproducible workflow for comparing mitigation strategies and establishing physical deployment frontiers.

2 Related Work

2.1 Hardware-Aware Training and Robustness

Hardware-aware training (HAT) mitigates analog non-idealities by injecting hardware perturbations into the forward path during optimization [13]. Although related in spirit to quantization-aware training [14] and to domain randomization in sim-to-real transfer [15], HAT for CIM faces an additional difficulty: device-to-device (D2D) mismatch is spatially structured and fixed for a given hardware instance. It is therefore different from independent thermal or sampling noise. Ensemble HAT resamples the static D2D mask at each training epoch, improving zero-shot transfer to fresh hardware instances. The connection to domain randomization is deliberate, but the analogy is incomplete because hardware-induced perturbations decompose into D2D mismatch—spatially structured and fixed for a given fabricated instance—and cycle-to-cycle (C2C) noise—stochastic per-read with zero mean.

2.2 Organic Neuromorphic and Optoelectronic Devices

Organic synaptic and memristive devices offer optical sensitivity, multilevel conductance tuning, and flexible-substrate compatibility for neuromorphic hardware [6;8;16;17]. Recent demonstrations include multilevel memory [7;11], long retention tails [10], and integrated optoelectronic arrays [18–21]. Array-level studies are beginning to treat active-matrix addressing, spatial optical response, and crosstalk as system constraints rather than purely materials observations [22–24], while temperature-resilient organic synapses and high-temperature synaptic phototransistors suggest that thermal stability is also becoming deployment-relevant [25;26].

2.3 CIM Simulation Frameworks and Hybrid Mapping

System-level CIM simulators such as DNN+NeuroSim established an effective template for connecting device assumptions, peripheral-circuit costs, and neural-network accuracy [2]. MemTorch showed how memristive non-idealities can be embedded into PyTorch workflows for end-to-end evaluation [9], AIHWKIT exposed analog HAT and inference simulation in a practical mixed-signal software stack [4], and CrossSim provides a GPU-accelerated crossbar-accuracy workflow with parasitic resistance, ADC, drift, and lookup-table-based training models [5]. These mature simulators were developed primarily around inorganic resistive memories, and they do not directly expose all profile-level effects of interest, such as inverse-gamma optoelectronic photoresponse or

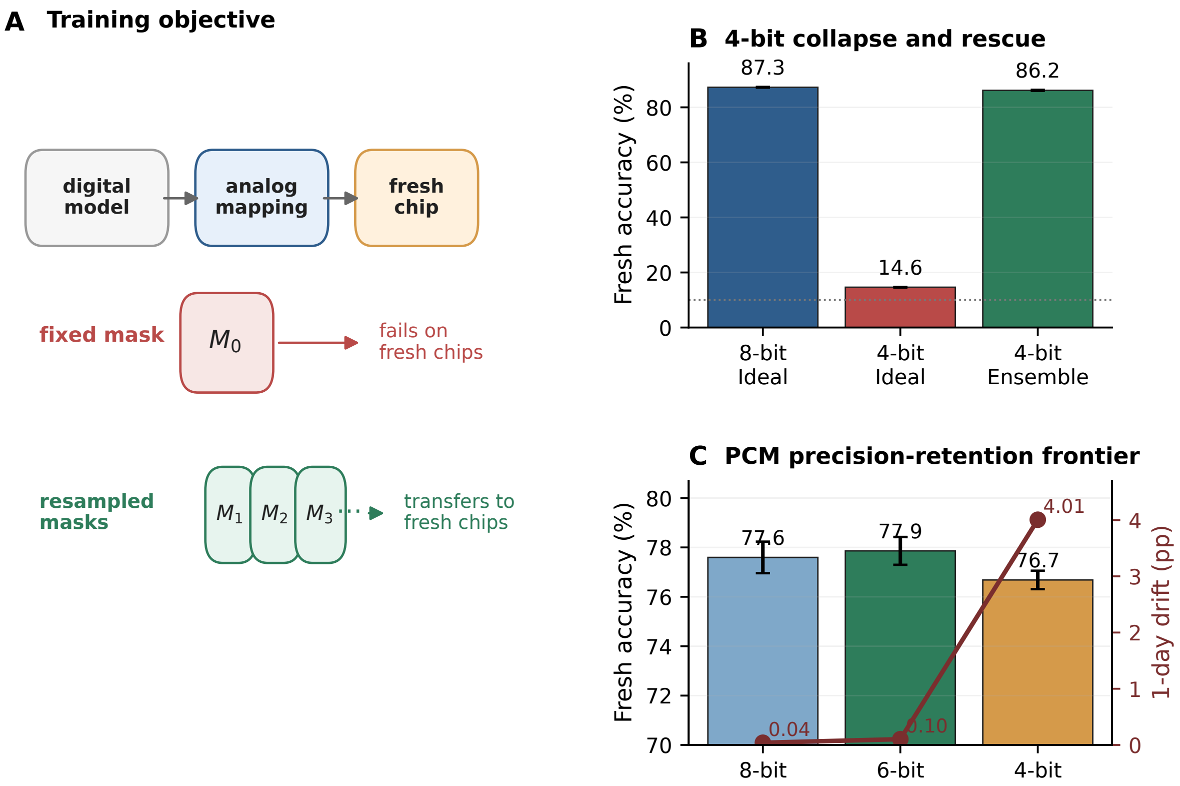

Figure 1: Paper-1 experimental spine. (A) Standard fixed-mask HAT optimizes one device instance, whereas Ensemble HAT trains over resampled D2D masks. (B) In the idealized AIHWKit baseline, 4-bit fresh-instance accuracy collapses while Ensemble HAT recovers it. (C) Under PCM UnitCell simulation, 4/6/8-bit models all train, but 4-bit has larger one-day drift and 6-bit is the best tested precision-retention midpoint.

organic-specific double-exponential retention, as first-class device primitives. Our behavioral profile-driven approach therefore serves as a profile-level extension and deployment-risk ranking layer rather than as a replacement for established CIM toolkits.

3 Results

3.1 The Failure of Pure Quantization and Algorithmic Rescue

We first measure the behavior of the tested AIHWKit IdealDevice quantization/noise baseline. At 8-bit precision the baseline remains stable, achieving 87.28 $\pm$ 0.13% fresh-instance accuracy; at 4-bit precision it collapses to 14.64 $\pm$ 0.11%. This indicates that standard fixed-instance hardware-aware training fails to generalize under fresh hardware instance shifts at extreme low precision in this tested regime.

To address this, we apply Ensemble Hardware-Aware Training (HAT). By resampling the device-to-device (D2D) variation patterns during the forward pass, Ensemble HAT rescues the tested 4-bit pure-quantization regime where the AIHWKit IdealDevice baseline collapses. The model recovers to 86.16 $\pm$ 0.19% fresh-instance accuracy, effectively closing the gap with the 8-bit baseline and demonstrating robust cross-instance transfer.

3.2 Realistic PCM Training and the Precision-Retention Frontier

While Ensemble HAT addresses the algorithmic generalization problem in a pure-quantization context, realistic analog deployment must account for complex device physics, particularly conductance drift. We evaluate our approach using the `PCMPresetUnitCell` model in AIHWKit, which incorporates non-linear update physics, finite dynamic range, and time-dependent drift.

Under the tested PCM UnitCell simulation regime, we observe 4-bit and 6-bit convergence where the pure quantization baseline collapses. As shown in Figure 1 and Table 1, training is viable across 4-bit, 6-bit, and 8-bit precisions. However, the results reveal a precision-retention deployment frontier for the tested PCM simulation regime.

Table 1: PCM precision ladder and deployment frontier. Values are 3-seed means. Source and fresh columns are accuracy (%); drift is the 0 s to 1 day drop in percentage points. All 6-bit PCM statistics use the matched full 100-epoch protocol because one seed recovered late after a false early plateau.

Precision	Source best (%)	Fresh mean (%)	1-day drift (pp)	Deployment role
8-bit PCM	77.64 ± 0.68	77.60 ± 0.64	0.04	Drift-flat reference
6-bit PCM	77.88 ± 0.58	77.86 ± 0.56	0.10	Pareto midpoint
4-bit PCM	76.71 ± 0.46	76.68 ± 0.37	4.01	Drift-limited

3.3 Deployment Guidance

The precision ladder clearly separates the algorithmic challenge of cross-instance robustness from the device-level precision-retention tradeoff.

8-bit PCM serves as a highly stable reference, remaining effectively drift-flat over the tested 24-hour retention window. Conversely, while 4-bit PCM achieves significant memory compression and remains highly trainable ($76.68 \pm 0.37\%$ fresh accuracy), it incurs a measurable time-dependent drift, dropping by $\sim$4.01 pp over one day.

In this sweep, 6-bit provides the best observed Pareto midpoint between model compression and long-term inference reliability in the tested PCM UnitCell setting. It maintains 8-bit-like drift stability (dropping only 0.10 pp over one day) while achieving a fresh accuracy of $77.86 \pm 0.56\%$, comparable to or slightly exceeding the 8-bit aggregate.

3.4 Future Directions

Extending the same hardware-aware training principles to memory-bound inference components such as analog KV-cache is an important future direction.

4 Discussion

4.1 Diagnosis: hardware-instance overfitting as the leading tested failure mode

Once hardware-aware training (HAT) and scale recovery are available, standard cycle-to-cycle (C2C) and device-to-device (D2D) variability do not dominate the error budget. Three constraints dominate instead.

A quantitative decomposition supports this ordering: designers should first secure 6-bit readout, then invest the remaining error budget in reducing device-to-device mismatch.

The first is ADC resolution. Below 6 bits, transformer inference degrades substantially, indicating that conductance control remains gated by readout precision.

The second is hardware-instance overfitting. The collapse of V4 under fresh-instance transfer (10.00%) shows that standard HAT readily fits a single fixed D2D realization. Replacing that fixed-mask training recipe with the mismatch-distributed objective in Eq. 2 raises fresh-instance accuracy to $86.16 \pm 0.19\%$ across three training seeds. Mechanistically, Ensemble HAT changes the mismatch map across epochs rather than merely perturbing one nominal instance.

A residual gap to the 98.06% digital baseline persists; part may reflect regularization from per-epoch D2D resampling. Capacity-aware Ensemble HAT schedules remain an open problem.

The third is the scale-masking regime in V2. When the recovered-weight law (Eq. 1) maps analog perturbations below the 4-bit quantization step, many realizations remain in the same recovered weight bin. This protection disappears under state-dependent noise.

4.2 Treatment: Ensemble HAT mitigation across scenarios

The main-text evidence supports a narrow but useful treatment claim. Ensemble HAT is appropriate when the leading observed failure mode is fresh-instance transfer under an idealized low-precision noise model: in that regime, resampling D2D mismatch during training converts a 4-bit collapse into 8-bit-like fresh-instance accuracy. It should not be read as a complete device-physics solution. Once the training objective is robust to fresh instances, realistic PCM simulation exposes a separate precision-retention frontier: the 4-bit model remains trainable but drifts over the tested 24-hour window, whereas the 6-bit and 8-bit models are effectively drift-flat. The practical rule is therefore sequential in this framework: first address cross-instance generalization with distribution-level HAT, then choose the analog precision from the retention frontier.

4.3 Mechanism: theory and empirical landscape

Supplementary Note S-Theory derives that the per-epoch resampling objective is, to second order, equivalent to a weighted gradient-L_2 regularizer (analogous to dropout-as-L_2, Wager et al. 2013) and, structurally, to a stochastic Sharpness-Aware Minimization along the device-mismatch direction (analogous to SAM, Foret et al. 2021). A PAC-Bayes argument provides a generalization-bound rationale for why fresh hardware-instance accuracy tracks the training-distribution average.

Empirical diagnostics support this interpretation. Loss-landscape interpolation from the training D2D mask toward fresh masks (Supplementary Note S-Mechanism, E2) shows that Ensemble HAT maintains $86.16 \pm 0.19\%$ accuracy when evaluated on a fresh mismatch map, whereas Standard HAT collapses to 10.00%—a gap of $\sim$76 pp. Under extrapolated mismatch ($3\times$ amplitude), Ensemble HAT still outperforms Standard HAT by 17.5 pp, indicating that the robustness extends beyond the training distribution. Per-layer sensitivity analysis (Supplementary Note S-Mechanism, E4) is retained as supplementary context for earlier severe-nonlinear-write diagnostics; it is not used here as a main-text localization claim under the revised precision-ladder narrative.

Full-parameter-space Hessian spectra (Supplementary Note S-Mechanism, E1) reveal a more nuanced picture: Standard HAT exhibits lower top eigenvalues at its training mask than Ensemble HAT, yet this “flatness” is mask-specific and does not transfer to fresh instances. The relevant robustness metric is therefore D2D-directional sensitivity rather than global curvature, aligning with the theoretical regularizer derived in Supplementary Note S-Theory.

4.4 Implications: design rules for deployment

Box 1: Design rules for analog CIM transformer deployment

1. **Secure $\geq$6-bit ADC readout first.** Below 6-bit, transformer inference degrades by $\sim$7 pp per row across all tested D2D levels.
2. **Then minimize device-to-device variability.** Within the operational envelope ($\geq$6-bit, $\sigma_{\text{D2D}} \leq$ 15%), D2D explains 92.2% of residual accuracy variance.
3. **Train with mismatch-distributed objective (Ensemble HAT), not fixed-mask.** Standard fixed-mask HAT collapses on fresh hardware instances (10.00%); Ensemble HAT recovers $86.16 \pm 0.19\%$ canonical across three seeds and $88.53 \pm 0.08\%$ on the literature-calibrated reference.
4. **Per-epoch D2D resampling, not per-batch.** Empirical ablation: epoch-level reaches 88.41%; per-batch only 86.16%; fixed 87.18%.
5. **Apply scale-recovery calibration after readout.** Without it, the analog perturbation is no longer bounded below the LSB and the recovered-weight protection vanishes.
6. **Compensate front-end sublinear photoresponse via inverse-gamma preprocessing.** Restores up to +5.8 pp at $\gamma_{\text{phys}} = 2.0$.
7. **Calibrate against measured D2D distribution when available.** Literature priors are an upper-bound proxy; deployment risk should be re-assessed once fabricated array statistics are in hand (see Supp Note S-HW).

The design rules in Box 1 are intentionally conservative: each threshold is chosen at the point where the accuracy–cost trade-off begins to degrade rapidly, not at the theoretical optimum. The 6-bit ADC threshold is the *cliff* observed in the iso-accuracy sweep, not the minimum ADC that could work under ideal calibration. The 15% D2D ceiling marks the boundary beyond which rank ordering between HAT variants destabilizes. Taken together, the rules frame deployment as sequential thresholding—readout first, then mismatch, then objective.

4.5 Comparison to established analog HAT primitives

At realistic deployment precision (4-bit, the canonical setting of our main analysis), we find that AIHWKit[4] per-batch noise injection achieves cross-instance robustness at 8-bit ($87.28 \pm 0.13\%$, fresh-instance) but collapses at 4-bit ($14.64 \pm 0.11\%$, fresh-instance). Ensemble HAT, by contrast, maintains $86.16 \pm 0.19\%$ at 4-bit (canonical three-seed aggregate). The 8-bit AIHWKit result is statistically comparable to the 4-bit Ensemble HAT result, indicating parity at higher precision. The advantage of Ensemble HAT in this comparison is therefore specific to the 4-bit regime—the practically relevant operating point where per-batch noise injection alone does not overcome mismatch variability. The collapse at this precision supports the importance of epoch-level D2D resampling as a structural analogue of distribution-matching objectives. When transitioning from idealized noise baselines to realistic phase-change memory simulation, the device model introduces a precision-dependent deployment frontier. Under the tested PCM UnitCell regime, 4-bit, 6-bit, and 8-bit models all converge, but retention behavior is precision dependent. The degradation is not uniform: 8-bit and 6-bit models remain drift-flat over the tested retention window, whereas the 4-bit model incurs a measurable 4.01 pp time-dependent drift. This frames the physical precision-retention tradeoff as a distinct engineering constraint alongside the algorithmic challenge of cross-instance generalization, positioning 6-bit as the best observed Pareto midpoint in the tested sweep.

4.6 Limitations and outlook

Scope and limitations. The conclusions above should be read against the boundaries of the present simulation stack. All experiments rest on device profiles calibrated from literature values or a single fabrication batch, so the accuracy projections do not extend to cross-batch variability, temperature drift, or aging. The severity of analog-induced degradation also scales with task complexity and data scarcity, which remains a fundamental limitation.

The framework is intentionally **behavioral, not circuit-accurate**: we simulate deployment-relevant non-idealities (quantization, D2D/C2C variability, ADC precision, write nonlinearity, retention) but defer SPICE-level phenomena (parasitic IR drop, sneak paths, pulse-faithful programming) to follow-on work. This scope choice mirrors established analog-CIM tooling[4], where behavioral fidelity suffices to **rank deployment risks** pre-fabrication. The profile-agnostic ingest interface accepts any measured device statistics (Supplementary Note S-HW); hardware-calibrated re-evaluation is anticipated once partner-array measurements are released.

The energy estimates are **first-order analytical projections** derived from literature-proxy per-operation constants ($E_{\text{cell}} = 100$ fJ, $E_{\text{ADC},8b} = 25$ fJ, $E_{\text{DAC},8b} = 30$ fJ; Section 5.5). They bound deployment risk to within an order of magnitude rather than predicting silicon performance. We deliberately avoid silicon-grade energy claims because the framework's contribution is **risk-ranking under realistic device profiles**, not chip-level optimization. A complete energy validation depends on partner-array characterization not yet available.

The present evaluation focuses on CIFAR-10/100 and Flowers-102 to enable systematic non-ideality and HAT-method ablation across three backbones within a tractable compute budget. **ImageNet-scale extension is the subject of an in-progress cross-architecture sweep** (ViT-Small/16, DeiT-Small/16 on TinyImageNet) on a separate compute infrastructure; preliminary results will appear in an extended companion study or supplementary erratum. The ablation matrix presented here intentionally trades dataset breadth for non-ideality coverage, motivating dataset-scale priorities for hardware validation.

Outlook. Three extensions follow naturally. First, measured-array calibration should replace the present UnitCell presets with device-specific noise and drift parameters and test whether the 6-bit Pareto midpoint persists. Second, cross-architecture validation on ViT-Small and DeiT-Small should determine whether the same robustness pattern holds beyond the Tiny-ViT setting before being elevated to a main-text claim. Third, memory-bound inference components such as analog KV-cache form a separate follow-on direction that should be evaluated with parity checks, held-out perplexity, and explicit train/eval noise separation before being merged into this framework. These extensions do not alter the present conclusion: the paper-1 evidence separates the algorithmic need for cross-instance resampling from the physical precision-retention frontier of PCM deployment.

5 Methodology

5.1 Hybrid Analog/Digital Deployment

We adopt a hybrid analog/digital inference stack: dense linear operators execute on differential-pair crossbars; control-heavy operators remain digital. Concretely, the analog partition contains patch embeddings, query/key/value (Q/K/V) and output projections, and feed-forward layers; the digital partition contains softmax, layer normalization, GELU activation, positional encoding, residual additions, and the class token, following established practice in heterogeneous CIM accelerators[27–31].

Under this mapping, approximately 88% of parameters execute on analog arrays while roughly 60% of estimated energy remains digital, largely because attention operations stay in the digital domain—a pattern consistent with recent heterogeneous CIM accelerators for Vision Transformers[28–31].

5.2 Modeling Physical Non-Idealities

The framework injects quantization, cycle-to-cycle (C2C) and device-to-device (D2D) variability, ADC distortion, retention drift, and nonlinear-write surrogates directly into the forward path. Each analog-mapped weight is decomposed into positive and negative branches, mapped to the conductance window $[G_{\min}, G_{\max}]$, and quantized to n_{states} levels via a straight-through estimator (STE) quantizer $Q_n(\cdot)$. Differential readout recovers $G_{\text{eff}} = \tilde{G}^+ - \tilde{G}^-$, and the downstream digital weight is

$$\hat{W}_{ij} = s_\ell \, G_{\text{eff},ij}, \qquad s_\ell = \begin{cases} \dfrac{w_{\max}}{G_{\max} - G_{\min}}, & \text{standard calibration} \\ \dfrac{w_{\max}}{\max_{i,j} |G_{\text{eff},ij}| + \epsilon}, & \text{retention-time recalibration} \end{cases} \tag{1}$$

with layer index ℓ omitted. Unless otherwise stated, we use the standard-calibration branch; the retention-recalibration branch is used only for the V8 retention-drift experiments. The recovered weight step is $\Delta_{\hat{W}} = s_\ell (G_{\max} - G_{\min})/(n_{\text{states}} - 1)$, so perturbations smaller than half a step are masked. The differential-pair scheme suppresses common-mode noise[32;33].

Hardware-aware training (HAT) injects forward perturbations while back-propagating through a straight-through estimator. Standard HAT fixes one D2D mismatch field M_0 during training.

Ensemble HAT as distribution matching. Ensemble HAT treats the mismatch map as a random variable resampled each epoch:

$$\theta^\star_{\text{ens}} = \arg\min_\theta \; \mathbb{E}_{M \sim \mathcal{D}_{\text{D2D}}} \left[\frac{1}{N} \sum_{n=1}^{N} \ell\big(f(x_n; \theta, M), y_n\big) \right], \tag{2}$$

where the expectation is Monte-Carlo estimated with one fresh draw $M^{(e)} \sim \mathcal{D}_{\text{D2D}}$ held constant across all mini-batches in epoch e. Expanding Eq. 2 (Supplementary Note S-Theory) reveals an implicit Fisher-weighted gradient-squared regularizer that pushes the optimizer toward flat regions in the M-direction, explaining the fresh-instance transferability without extra hyperparameters. Epoch-level (not per-batch) resampling is load-bearing, giving the optimizer enough consistent signal to fit each instance before resampling[15]. Fresh-instance transfer is then evaluated on unseen i.i.d. D2D draws held constant for each evaluation pass. For a single checkpoint, the reported fresh-instance spread is the standard deviation across ten fresh-instance means (each averaged over five forward passes); for the R10A reproducibility headline, the uncertainty is the sample standard deviation across three seed-level fresh-instance means.

5.3 Training Protocol

All Tiny-ViT experiments use AdamW ($lr = 5 \times 10^{-4}$, weight decay 0.05), 100 epochs, batch size 64, cosine annealing ($T_{\max} = 100$). Images are resized to 224×224 with standard augmentation. The default analog resolution is $n_{\text{states}} = 16$ levels. V1 is digital; V2 is hybrid without noise; V3 is hybrid with fixed D2D ($\sigma_{\text{D2D}} = 10\%$); V4–V7 use full HAT with epoch-resampled D2D and per-forward C2C ($\sigma_{\text{C2C}} = 5\%$ or 10%). V6 adds the photoresponse front end; V7 is a legacy retention-aware model excluded from canonical claims; V8 is the corrected retention-aware follow-up used for retention checks.

Nonlinear-write and retention surrogates are anchored to measured DNTT transients[34]. Nonlinear write modifies the STE backward pass by scaling the straight-through gradient with a state-dependent factor proportional to the present conductance magnitude raised to $NL - 1$[1];

[1]The absence of an explicit NL prefactor is intentional: this is a behavioral proxy for position-dependent update difficulty, not the strict derivative of a power-law $f(G) = G^{NL}$.

Table 2: Notation used for the Tiny-ViT experiment family in the main text.

ID	Meaning
V1	FP32 digital baseline
V2	Quantized hybrid control without analog noise
V3	Standard training with a fixed D2D realization; noisy deployment baseline
V4	Uniform-noise hardware-aware training result used as the canonical deployment model
V5	Pessimistic robustness stress test with larger uniform noise
V6	V4 plus physical front-end compensation
V7	Legacy retention-aware model, excluded from the canonical claims
V8	Corrected retention-aware follow-up used only for retention checks

ideal write is recovered when $NL_{\text{LTP}} = 1$ (long-term potentiation) and $|NL_{\text{LTD}}| = 1$ (long-term depression). The optional optoelectronic front-end model (V6) applies inverse-gamma preprocessing, photocurrent shot-noise injection, and per-sample renormalization; the explicit response equations are kept in the Supplementary Information because V6 is not part of the main canonical claim.

5.4 Profile-Driven Substitution Interface

The profile interface packages technology-specific assumptions into a replaceable JSON parameter bundle. Deployment holds the digital backbone and training recipe fixed while substituting the profile, enabling literature priors today and measured-device calibration later without code changes.

5.5 Sensitivity and Energy Metrics

For the joint D2D–ADC sweep, first-order factor importance is quantified with Sobol indices estimated from the 7×9 grid of Monte Carlo means. Energy is estimated via a first-order analytical model (Supplementary Methods) from operator counts and literature-proxy constants ($E_{\text{cell}} = 100$ fJ, $E_{\text{ADC},8b} = 25$ fJ, $E_{\text{DAC},8b} = 30$ fJ); these are analytical placeholders for relative comparison, not measured silicon values.

6 Experimental Setup

We evaluate ResNet-18 on CIFAR-10 as an entry validation platform, ConvNeXt-Tiny trained from scratch as the higher-capacity CNN testbed, and Tiny-ViT-5M fine-tuned from ImageNet as the deployment-oriented transformer setting. CIFAR-10, CIFAR-100, and Flowers-102 provide a progression from easier to harder classification settings and from relatively data-rich to data-limited regimes.

The experiments fall into three groups: ResNet baseline quantization and front-end trends, ConvNeXt retention and proportional-noise studies, and Tiny-ViT hardware-aware training together with physical-extension tests such as nonlinear write and fresh-instance transfer. Detailed optimization settings and the Monte Carlo evaluation protocol are summarized in the Supplementary Information.

For ConvNeXt, C1–C4 denote the analogous FP32, quantized/no-noise, noisy-baseline, and hardware-aware training runs; C4 also anchors the proportional-noise and retention follow-up studies described in the Supplementary Information.

Parameter provenance is summarized in the Supplementary Information, together with the optimization settings and evaluation conditions used for the reported runs. The present study therefore targets reproducible reruns through archived configurations, logs, and model lineage. We also emphasize that the framework operates at the simulation level: all device parameters are literature-derived or proxy-calibrated rather than extracted from fabricated devices under direct measurement. Validation against physical hardware remains for future work, and the quantitative results below should be interpreted as comparative risk rankings rather than absolute deployment predictions.

7 Conclusion

We presented a behavioral simulation framework for analog compute-in-memory inference that disentangles algorithmic generalization failures from physical precision-retention constraints. By comparing ideal pure-quantization baselines against realistic phase-change memory (PCM) physics, the framework serves as a practical decision aid for establishing deployment frontiers in low-precision analog hardware.

Several observations stand out. At extreme low precision (4-bit), fixed hardware-instance transfer is the most prominent simulated deployment risk observed in our matrix: naive analog deployment collapses to 14.64%, whereas Ensemble Hardware-Aware Training (HAT) resamples mismatch masks to rescue convergence, recovering fresh-instance accuracy to 86.16±0.19%. Under the tested PCM UnitCell simulation regime, convergence is observed across the 4/6/8-bit precision ladder where pure quantization fails. However, this convergence exposes a precision-retention tradeoff. While 8-bit PCM is drift-flat and 4-bit PCM reaches the most aggressive tested compression at the cost of measurable time-dependent drift, 6-bit emerges as the best observed Pareto midpoint in this sweep, maintaining 8-bit-like stability alongside robust fresh accuracy.

Ultimately, this framework clarifies that standard fixed-mask training is insufficient for extreme low-precision analog deployment. Ensemble HAT resolves this algorithmic bottleneck, allowing system designers to navigate the subsequent physical limits of the analog precision-retention frontier. Extending these hardware-aware training principles to memory-bound inference components, such as analog KV-cache, presents a compelling future trajectory for large-scale deployment.

Data Availability

All datasets used in this study (CIFAR-10, CIFAR-100, and Flowers-102) are publicly available through standard PyTorch torchvision and HuggingFace dataset interfaces. The source data underlying the main-text and Supplementary figures and tables are packaged in the submission-matched source-data archive together with the corresponding manifests, logs, and figure-generation inputs. A reproducibility archive containing the code snapshot, source data, SHA-256 manifest, and `CITATION.cff` file has been staged for Zenodo deposition and will be assigned a DOI at acceptance.

Code Availability

The simulation framework, training and evaluation scripts, device-profile utilities, and plotting code will be provided to editors and reviewers through a reviewer-accessible archive at submission. A curated public release will be made available at https://github.com/Leslie360/HAT upon acceptance. The current repository is distributed under the Apache-2.0 licence, and the staged Zenodo archive will be updated with the final public DOI at acceptance.

Competing Interests

The authors declare no competing interests.

References

[1] Mark Horowitz. Computing's energy problem (and what we can do about it). In *IEEE International Solid-State Circuits Conference (ISSCC)*, pages 10–14, 2014. doi: 10.1109/ISSCC.2014.6757323. URL https://doi.org/10.1109/ISSCC.2014.6757323.

[2] Xiaochen Peng, Shanshi Huang, Hongwu Jiang, Anni Lu, and Shimeng Yu. DNN+NeuroSim v2.0: An end-to-end benchmarking framework for compute-in-memory accelerators for on-chip training. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 40(11):2306–2319, 2021. doi: 10.1109/TCAD.2020.3043731. URL https://doi.org/10.1109/TCAD.2020.3043731.

[3] Mario Lanza et al. The growing memristor industry. *Nature*, 640:613–622, 2025. doi: 10.1038/s41586-025-08733-5.

[4] Malte J. Rasch, Diego Moreda, Tayfun Gokmen, Manuel Le Gallo, Fabio Carta, Cindy Goldberg, Kaoutar El Maghraoui, Abu Sebastian, and Vijay Narayanan. A flexible and fast PyTorch toolkit for simulating training and inference on analog crossbar arrays, 2021. URL https://arxiv.org/abs/2104.02184.

[5] Sandia National Laboratories. Crosssim: Sandia's simulator for analog ai accelerators. Technical report, Sandia National Laboratories, 2024. URL https://doi.org/10.2172/2585829.

[6] Yunchao Xu, Yuan He, Dongyong Shan, Biao Zeng, and Qian-Xi Ni. Emerging artificial synaptic devices based on organic semiconductors: Molecular design, structure and applications. *ACS Applied Materials & Interfaces*, 2025. doi: 10.1021/acsami.4c17455. URL https://doi.org/10.1021/acsami.4c17455.

[7] Haotian Guo, Jing Guo, Yujing Wang, Hezhen Wang, Simin Cheng, Zehao Wang, Qian Miao, and Xiaomin Xu. An organic optoelectronic synapse with multilevel memory enabled by gate modulation. *ACS Applied Materials & Interfaces*, 2024. doi: 10.1021/acsami.3c19624. URL https://doi.org/10.1021/acsami.3c19624.

[8] Li Zhang, Wei Chen, and Jing Xu. Recent progress in organic optoelectronic synaptic devices. *Photonics*, 12(5):435, 2025. doi: 10.3390/photonics12050435. URL https://doi.org/10.3390/photonics12050435.

[9] Corey Lammie, Wei Xiang, Bernabé Linares-Barranco, and Mostafa Rahimi Azghadi. Memtorch: An open-source simulation framework for memristive deep learning systems. *Neurocomputing*, 485:124–133, 2022. doi: 10.1016/j.neucom.2022.02.043. URL https://doi.org/10.1016/j.neucom.2022.02.043.

[10] Jianmin Zeng, Xinhui Chen, Shuzhi Liu, Qilai Chen, and Gang Liu. Organic memristor with synaptic plasticity for neuromorphic computing applications. *Nanomaterials*, 13(5):803, 2023. doi: 10.3390/nano13050803. URL https://doi.org/10.3390/nano13050803.

[11] Uihoon Jung, Miseong Kim, Jaewon Jang, Jin-Hyuk Bae, In Man Kang, and Sin-Hyung Lee. Formation of cluster-structured metallic filaments in organic memristors for wearable neuromorphic systems with bio-mimetic synaptic weight distributions. *Advanced Science*, 11(9):2307494, 2024. doi: 10.1002/advs.202307494. URL https://doi.org/10.1002/advs.202307494.

[12] Yu-Pu Lin, Christopher H. Bennett, Théo Cabaret, Damir Vodenicarevic, Djaafar Chabi, Damien Querlioz, Bruno Jousselme, Vincent Derycke, and Jacques-Olivier Klein. Physical

realization of a supervised learning system built with organic memristive synapses. *Scientific Reports*, 2016. doi: 10.1038/srep31932. URL https://doi.org/10.1038/srep31932.

[13] Vinay Joshi, Manuel Le Gallo, Simon Haefeli, Irem Boybat, S. R. Nandakumar, Christophe Piveteau, Martino Dazzi, Bipin Rajendran, Abu Sebastian, and Evangelos Eleftheriou. Accurate deep neural network inference using computational phase-change memory. *Nature Communications*, 11:2473, 2020. doi: 10.1038/s41467-020-16108-9. URL https://doi.org/10.1038/s41467-020-16108-9.

[14] Jungwook Choi, Zhuo Wang, Swagath Venkataramani, Pierce I-Jen Chuang, Vijayalakshmi Srinivasan, and Kailash Gopalakrishnan. PACT: Parameterized clipping activation for quantized neural networks. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1028–1037, 2018. doi: 10.48550/arXiv.1805.06085. URL http://proceedings.mlr.press/v80/choi18a.html.

[15] Josh Tobin, Rachel Fong, Alex Ray, Jonas Schneider, Wojciech Zaremba, and Pieter Abbeel. Domain randomization for transferring deep neural networks from simulation to the real world. In *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 23–30, 2017. doi: 10.1109/IROS.2017.8202133. URL https://doi.org/10.1109/IROS.2017.8202133.

[16] Pietro Belleri, Judith Pons i Tarrès, Iain McCulloch, Paul W. M. Blom, Zsolt M. Kovács-Vajna, Paschalis Gkoupidenis, and Fabrizio Torricelli. Unravelling the operation of organic artificial neurons for neuromorphic bioelectronics. *Nature Communications*, 15:5350, 2024. doi: 10.1038/s41467-024-49668-1. URL https://doi.org/10.1038/s41467-024-49668-1.

[17] Junpeng Ji, Dace Gao, Han-Yan Wu, Miao Xiong, Nevena Stajkovic, Claudia Latte Bovio, Chi-Yuan Yang, Francesca Santoro, Deyu Tu, and Simone Fabiano. Single-transistor organic electrochemical neurons. *Nature Communications*, 16:4334, 2025. doi: 10.1038/s41467-025-59587-4. URL https://doi.org/10.1038/s41467-025-59587-4.

[18] Lan Shen Hu, Marco Fattori, Winston Schilp, Roy Verbeek, Setareh Kazemzadeh, Yoeri van de Burgt, Auke Jisk Kronemeijer, Gerwin Gelinck, and Eugenio Cantatore. An energy-efficient solid-state organic device array for neuromorphic computing. *IEEE Transactions on Electron Devices*, 70(12):6520–6525, 2023. doi: 10.1109/TED.2023.3327947. URL https://doi.org/10.1109/TED.2023.3327947.

[19] Xu Liu, Shilei Dai, Yiyang Jin, Junyao Zhang, Ziyi Guo, Tongrui Sun, Li Li, Pu Guo, Huaiyu Gao, Haixia Liang, Shiqi Zhang, Lize Xiong, Yanmin Zhou, and Jia Huang. Near-infrared organic photoelectrochemical synaptic transistors by wafer-scale photolithography for neuromorphic visual system. *Nature Communications*, 17:197, 2026. doi: 10.1038/s41467-025-66891-6. URL https://doi.org/10.1038/s41467-025-66891-6. Published online 2025-12-08.

[20] Fanqing Zhang, Xinyi Shan, Zhenhua Ji, Yan Wang, Jing Zhao, and Zhongrui Wang. Large-scale high uniform optoelectronic synapses array for artificial visual neural network. *Microsystems & Nanoengineering*, 11:42, 2025. doi: 10.1038/s41378-024-00859-2. URL https://doi.org/10.1038/s41378-024-00859-2.

[21] Heyi Huang, Xiangpeng Liang, Yuyan Wang, Jianshi Tang, Yuankun Li, Yiwei Du, Wen Sun, Jianing Zhang, Peng Yao, Xing Mou, Feng Xu, Jinzhi Zhang, Yuyao Lu, Zhengwu Liu, Jianlin Wang, Zhixing Jiang, Ruofei Hu, Ze Wang, Qingtian Zhang, Bin Gao, Xuedong Bai, Lu Fang, Qionghai Dai, Huaxiang Yin, He Qian, and Huaqiang Wu. Fully integrated multi-mode optoelectronic memristor array for diversified in-sensor computing. *Nature*

Nanotechnology, 20(1):93–103, 2025. doi: 10.1038/s41565-024-01794-z. URL https://doi.org/10.1038/s41565-024-01794-z.

[22] Ying Liu, Biao Wang, Lei Wu, Lulu Huang, Lu Lin, Li Xiang, Dongqing Liu, Shiguo Zhang, Chenguang Zhu, and Yijie Tao. Artificial visual synaptic architecture with high-linearity light-modulated weight for optoelectronic neuromorphic computing. *ACS Applied Materials & Interfaces*, 2023. doi: 10.1021/acsami.3c11495. URL https://doi.org/10.1021/acsami.3c11495.

[23] Dingwei Li, Yitong Chen, Huihui Ren, Yingjie Tang, Siyu Zhang, Yan Wang, Lixiang Xing, Qi Huang, Lei Meng, and Bowen Zhu. An active-matrix synaptic phototransistor array for in-sensor spectral processing. *Advanced Science*, 11, 2024. doi: 10.1002/advs.202406401. URL https://doi.org/10.1002/advs.202406401.

[24] Tianqing Wan, Bangjie Shao, Sijie Ma, Yue Zhou, Qiao Li, and Yang Chai. In-sensor computing: Materials, devices, and integration technologies. *Advanced Materials*, 35(37): 2203830, 2023. doi: 10.1002/adma.202203830. URL https://doi.org/10.1002/adma.202203830.

[25] A. Melianas, T. J. Quill, G. LeCroy, Y. Tuchman, H. v. Loo, S. T. Keene, A. Giovannitti, H. R. Lee, I. P. Maria, I. McCulloch, and A. Salleo. Temperature-resilient solid-state organic artificial synapses for neuromorphic computing. *Science Advances*, 6:eabb2958, 2020. doi: 10.1126/sciadv.abb2958. URL https://doi.org/10.1126/sciadv.abb2958.

[26] Ziyi Guo, Junyao Zhang, Ben Yang, Li Li, Xu Liu, Yutong Xu, Yue Wu, Pu Guo, Tongrui Sun, Shilei Dai, Haixia Liang, Jun Wang, Yidong Zou, Lize Xiong, and Jia Huang. Organic high-temperature synaptic phototransistors for energy-efficient neuromorphic computing. *Advanced Materials*, 36:2310155, 2024. doi: 10.1002/adma.202310155. URL https://doi.org/10.1002/adma.202310155.

[27] Xueying Wu, Edward Hanson, Nansu Wang, Qilin Zheng, Xiaoxuan Yang, Huanrui Yang, Shiyu Li, Feng Cheng, Partha Pratim Pande, and Janardhan Rao Doppa. Block-wise mixed-precision quantization: Enabling high efficiency for practical ReRAM-based DNN accelerators. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2024. doi: 10.1109/TCAD.2024.3409193. URL https://doi.org/10.1109/TCAD.2024.3409193.

[28] Chenyu Wang, Zhen Dong, Daquan Zhou, Zhenhua Zhu, Yu Wang, Jiashi Feng, and Kurt Keutzer. EPIM: Efficient processing-in-memory accelerators based on epitome. In *Proceedings of the 61st ACM/IEEE Design Automation Conference (DAC)*, pages 1–6, 2024. doi: 10.1145/3649329.3657377. URL https://arxiv.org/abs/2311.07620.

[29] Mengke Ge, Junpeng Wang, Binhan Chen, Yingjian Zhong, Haitao Du, Song Chen, and Yi Kang. Allspark: Workload orchestration for visual transformers on processing in-memory systems. *IEEE Transactions on Computers*, 2025. doi: 10.1109/TC.2024.3483633. URL https://doi.org/10.1109/TC.2024.3483633.

[30] Cong Wang, Zexin Fu, Jiayi Huang, and Shanshi Huang. Hemlet: A heterogeneous compute-in-memory chiplet architecture for vision transformers with group-level parallelism, 2025. URL https://arxiv.org/abs/2511.15397.

[31] Shiwei Liu, Chen Mu, Hao Jiang, Yunzhengmao Wang, Jinshan Zhang, Feng Lin, Keji Zhou, Qi Liu, and Chixiao Chen. HARDSEA: Hybrid analog-ReRAM clustering and digital-SRAM in-memory computing accelerator for dynamic sparse self-attention in transformer. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, 32(2):269–282, 2024. doi: 10.1109/TVLSI.2023.3337777. URL https://doi.org/10.1109/TVLSI.2023.3337777.

[32] W. Wan, R. Kubendran, B. Gao, S. Joshi, P. Raina, H. Wu, G. Cauwenberghs, and H.-S. P. Wong. A voltage-mode sensing scheme with differential-row weight mapping for energy-efficient RRAM-based in-memory computing. In *2020 IEEE Symposium on VLSI Technology*, pages 1–2, 2020. doi: 10.1109/VLSITechnology18217.2020.9265066. URL https://doi.org/10.1109/VLSITechnology18217.2020.9265066.

[33] Xianggao Wang, Na Wei, Shifan Gao, Wenhao Wu, and Yi Zhao. Accuracy improvement with weight mapping strategy and output transformation for STT-MRAM-based computing-in-memory. *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits*, 10:75–81, 2024. doi: 10.1109/JXCDC.2024.3478360. URL https://doi.org/10.1109/JXCDC.2024.3478360.

[34] Tomas Vincze, Michal Hanic, Martin Berki, and Martin Weis. Simultaneous dual-plasticity organic synaptic transistor for neuromorphic computing. *Advanced Electronic Materials*, 12(1):e00515, 2026. doi: 10.1002/aelm.202500515. URL https://doi.org/10.1002/aelm.202500515. Published online 2025-12-03.